# Project Output

Python 3.14.5 (tags/v3.14.5:5607950, May 10 2026, 10:43:50) [MSC v.1944 64 bit (AMD64)] on win32

Enter "help" below or click "Help" above for more information.

= RESTART: E:/Real Lif Project In Data Analyst/Advance Numpy Retail Sales Business Analytics/Retail Sales Business Analytics.py

==========Part A Sales Analysis==========

======Units Sold Unit Price Adjusted For Discount======

[1800000  100000       0 2640000  450000  180000       0  540000 1120000

  270000   52500  400000  630000       0  456000]

======Order With Revenue Above 50000======

[201 204 209 213]

======Find Orders With Discount Greater Than 8 Percent======

[201 204 205 208 209 210 212 215]

======Calculate Average Revenue======

57034.333333333336

======Order Below Average Revenue======

[202 203 205 206 207 208 210 211 212 214 215]

==========Part B Productivity & Profit==========

======Calculate Profit Per Unit Sold======

[2000.         200.          45.        2250.         500.

  300.          29.09090909  820.        3200.         344.44444444

   54.28571429  475.        2071.42857143  250.         700.       ]

======Calculate Revenue Per Unit======

[13500.   760.   250. 18700.  2700.  1140.   180.  3960. 25760.  1350.

   285.  2300. 17100.   950.  3420.]

======Orders With Profit Above 8000======

[24000  5000  1800 18000  7500  9000  1600  8200 16000  6200  1900  9500

14500  5500  8400]

======Orders With More Than 20 Units Sold======

[202 203 206 207 211 212 214]

======Order With Profit Above 5000 and Discount Below 10%======

[201 202 205 206 209 210 212 213 214 215]

==========Part C Performance Analysis==========

======Calculate Average Profit======

9140.0

======The Highest Profit======

24000

======Lowest Revenue======

9900

======Median Revenue======

39600.0

======Standard Deviation Of Revenue======

52033.38962841285

==========Part D Order Classification==========

======Order Revenue High to Low======

['High Value' 'High Value' 'High Value' 'High Value' 'High Value'

 'High Value' 'Medium Value' 'High Value' 'High Value' 'High Value'

 'Medium Value' 'High Value' 'High Value' 'High Value' 'High Value']

======Classify Profit======

['Excellent' 'Good' 'Needs Review' 'Excellent' 'Good' 'Good'

 'Needs Review' 'Good' 'Excellent' 'Good' 'Needs Review' 'Good'

 'Excellent' 'Good' 'Good']

======Classify Discount======

['High Discount' 'Regular Discount' 'Regular Discount' 'High Discount'

 'High Discount' 'Regular Discount' 'Regular Discount' 'High Discount'

 'Regular Discount' 'High Discount' 'Regular Discount' 'Regular Discount'

'Regular Discount' 'Regular Discount' 'High Discount']

==========Part E Management Insights==========

======Identify High Revenue But Low Profit Orders======

[204 209 213]

======Identify High Discount Orders With Low Untis Sold======

[201 204 205 208 215]

======Orders with Strong Profit======

[201 206 208 212 215]

======Identify Orders Profit Low & Discount HighNeeding Review======

[201 203 204 205 207 208 210 211 215]

==========Part F Mathematical Analysis==========

======Round Profit Margin Percentage to 2 Decimal Places======

[24000 5000 1800 18000 7500 9000 1600 8200 16000 6200 1900 9500

14500 5500 8400]

======Calculate Square Root Of Units Sold======

[3.46410162 5. 6.32455532 2.82842712 3.87298335 5.47722558

7.41619849 3.16227766 2.23606798 4.24264069 5.91607978 4.47213595

2.64575131 4.69041576 3.46410162]

======Calculate Squared Revenue Values======

[26244000000 361000000 100000000 22380160000 1640250000 1169640000

98010000 1568160000 16589440000 590490000 99500625 2116000000

14328090000 436810000 1684281600]

======Calculate Absolute Difference Between Each Order's Revenue And Average Revenue======

[162000 19000 10000 149600 40500 34200 9900 39600 128800 24300

9975 46000 119700 20900 41040]

==========Part G Final Business Report==========

Net Revenue

[138000 14000 8200 131600 33000 25200 8300 31400 112800 18100

8075 36500 105200 15400 32640]

Profit Pr Unit Sold

[2000.  200.  45.  2250.  500.

 300.  29.09090909  820.  3200.  344.44444444

 54.28571429  475.  2071.42857143  250.  700.  ]

Profit Margin

[14.81481481 26.31578947 18.  12.03208556 18.51851852 26.31578947

 16.16161616 20.70707071 12.42236025 25.51440329 19.04761905 20.65217391

 12.11361738 26.31578947 20.46783626]

=====Count Revenue Values======

Count High Value: 13

Count Medium Values: 2

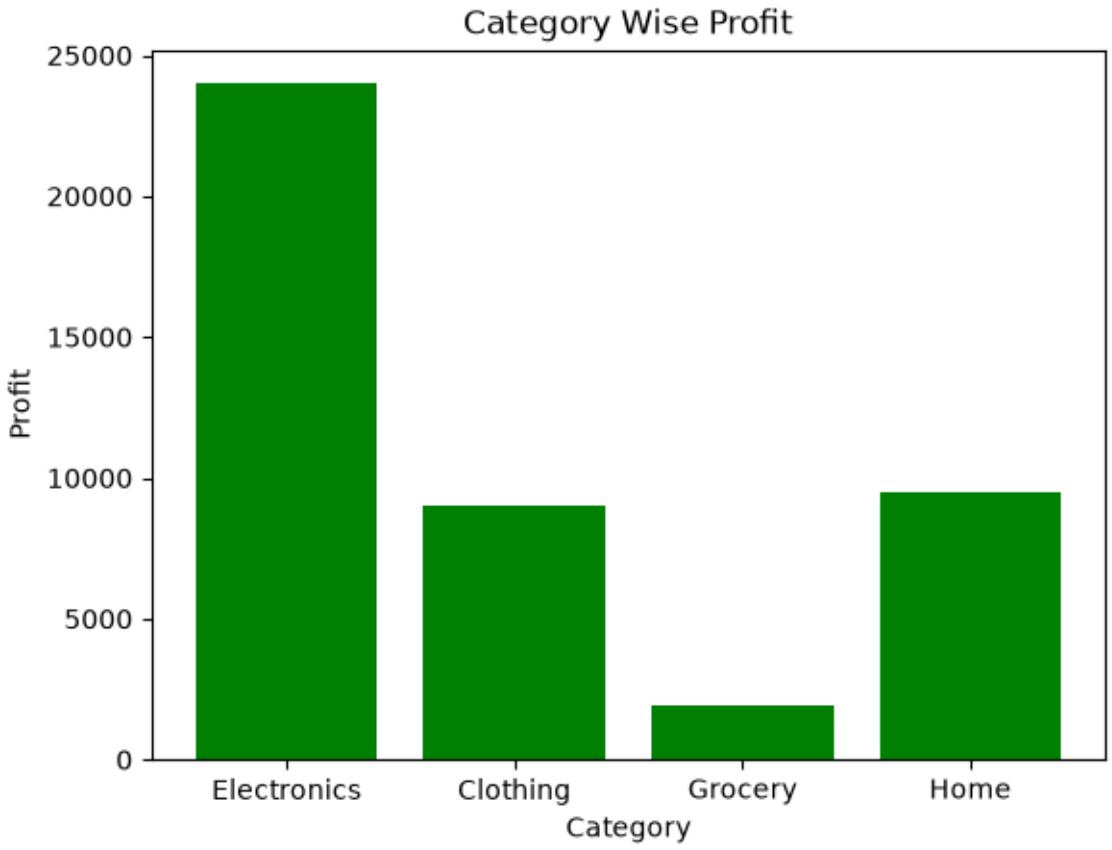
Count Low Value: 0

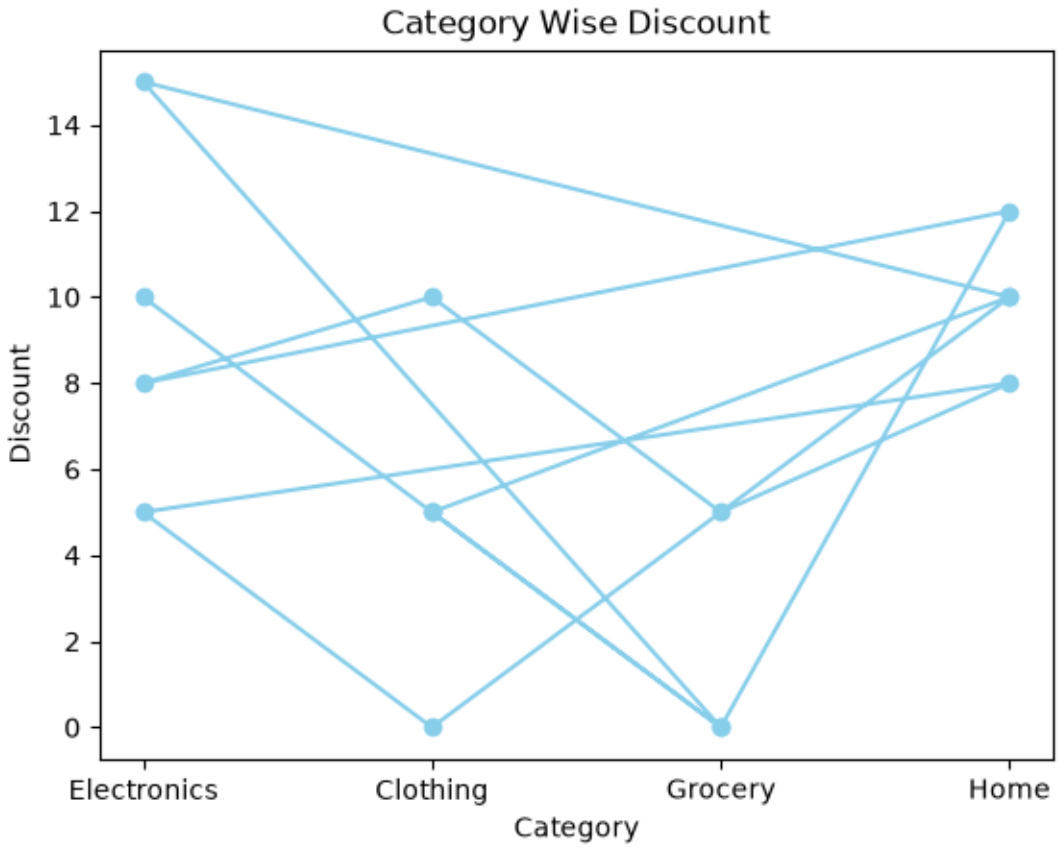
=====Find Total Revenue Total Profit======

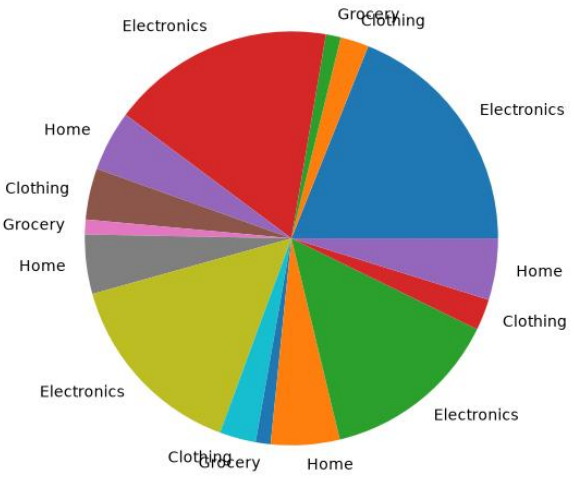
Total Revenue: 855515

Total Profit: 137100

==========Part H Chart==========




Category Wise Discount
Discount
14
12
10
8
6
4
2
0
Electronics
Clothing
Grocery
Home
Category


Electronics
Grocery
Clothing
Electronics
Home
Clothing
Grocery
Home
Home
Clothing
Electronics
Electronics
Clothing
Grocery
Home